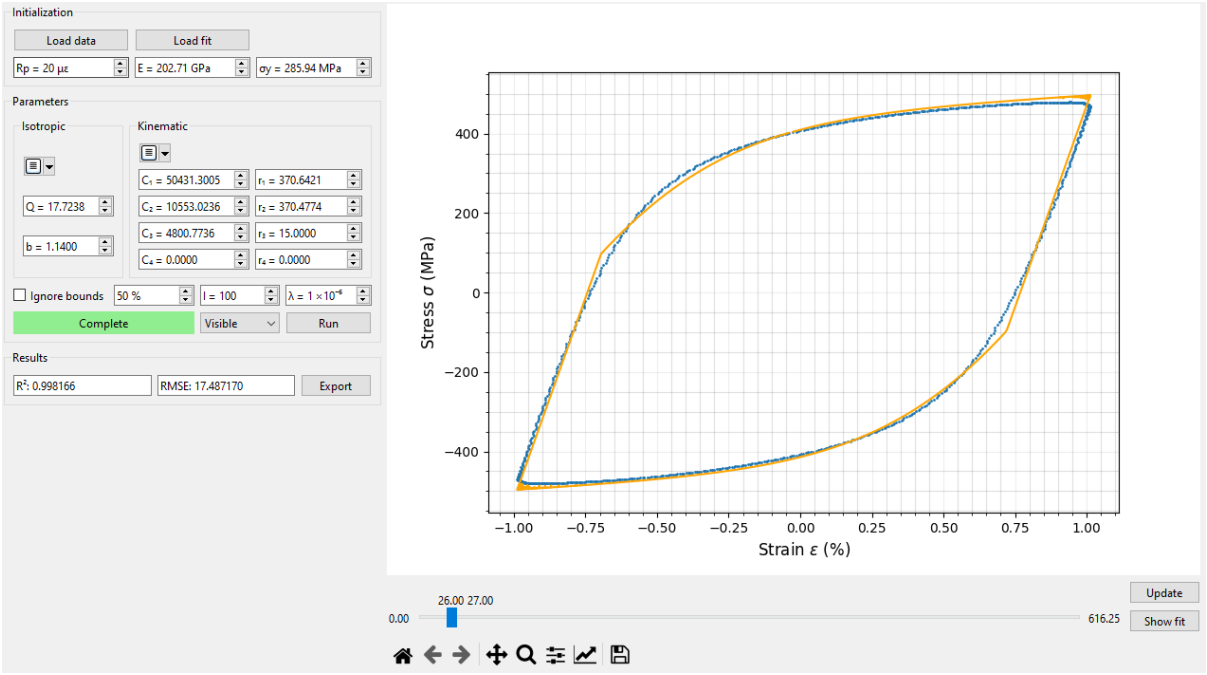
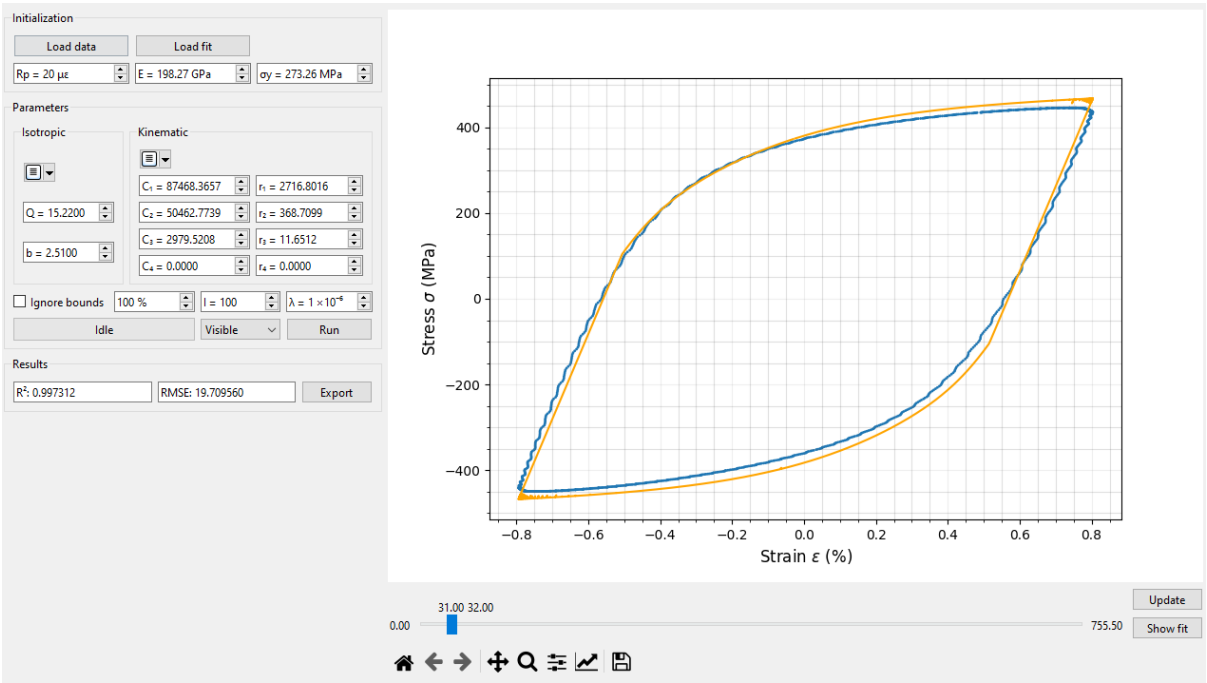
Sample	$\Delta\varepsilon(\%)$	*Saturation cycle starts at*	*Linear Parameter*		*Isotropic Parameter*	
			E(GPa)	σy (MPa)	Q (MPa)	b
A	1	26	202	283	11.46	1.31
A	0.8	31	198	273	15.22	2.51
A	0.6	32	181	199	36.42	3.10
B	1	22	207	229	27.86	1.29
B	0.8	29	202	221	28.94	7.53
B	0.6	21	200	211	39.92	1.51
C	1	26	193	241	25.54	1.08
C	0.8	32	190	233	53.89	1.10
C	0.6	21	186	223	56.49	1.95
H	1	26	203	363	24.55	1.15
H	0.8	32	201	347	30.31	2.52
H	0.6	33	200	318	75.87	5.77
H	0.4	36	159	270	150.00	10.00
V	1	22	171	323	12.11	1.10
V	0.8	30	168	315	19.30	1.98
V	0.6	32	163	300	27.08	5.70
V	0.4	35	159	254	36.05	10.00

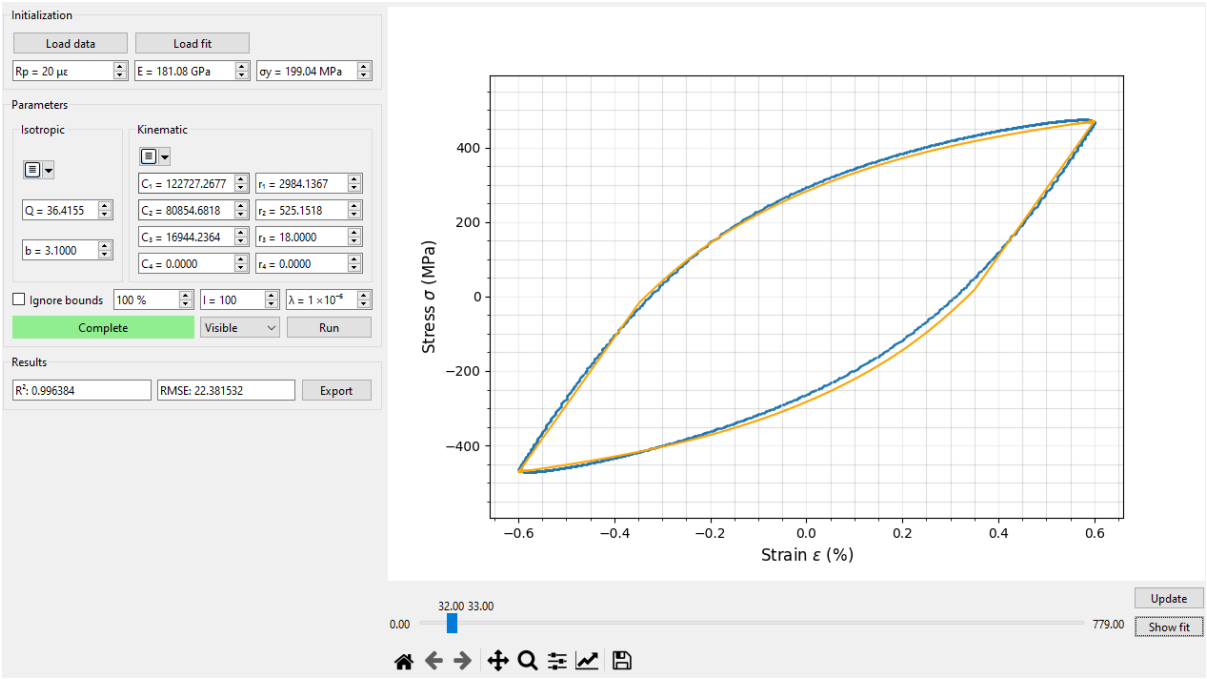
Sample	$\Delta\varepsilon(\%)$	*Kinematic parameters*					
		c1(MPa)	r1	c2(MPa)	r2	c3(MPa)	r3
A	1	70452	867	30568	281	2859	9
A	0.8	87468	2716	50462	368	2979	11
A	0.6	122727	2984	80854	525	16944	18
B	1	55075	1089	33620	209	3696	194
B	0.8	66501	1132	34171	268	3975	275
B	0.6	69278	1256	39153	288	4862	277
C	1	72285	3526	49169	386	19870	88
C	0.8	81478	3949	51854	501	28296	109
C	0.6	81341	7545	69822	566	32921	122
H	1	79416	481	8291	319	4530	113
H	0.8	88978	787	17933	346	11931	221
H	0.6	99379	1249	42750	881	15191	294
H	0.4	115250	3999	65911	1398	23624	599
V	1	34963	464	9182	159	7223	133
V	0.8	75233	590	10841	485	9531	218
V	0.6	95308	1244	25915	592	15597	397
V	0.4	123558	1156	56399	1347	28436	600


Initialization
Load data
Load fit
Rp = 20 με
E = 202.71 GPa
σy = 285.94 MPa
Parameters
Isotropic
Kinematic
Q = 17.7238
b = 1.1400
C1 = 50431.3005
r1 = 370.6421
C2 = 10553.0236
r2 = 370.4774
C3 = 4800.7736
r3 = 15.0000
C4 = 0.0000
r4 = 0.0000
Ignore bounds
50 %
l = 100
λ = 1×10⁻⁴
Complete
Visible
Run
Results
R²: 0.998166
RMSE: 17.487170
Export
Stress σ (MPa)
Strain ε (%)
26.00 27.00
0.00
616.25
Update
Show fit

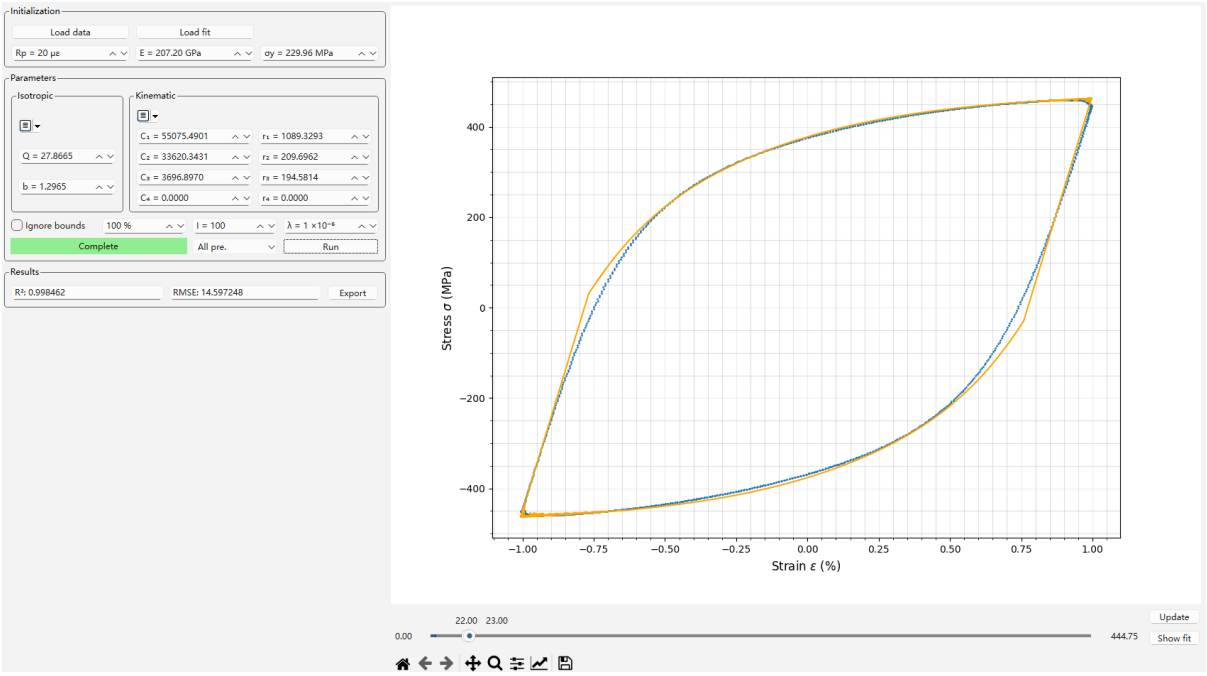
A_1am


Initialization
Load data
Load fit
Rp = 20 με
E = 198.27 GPa
σy = 273.26 MPa
Parameters
Isotropic
Kinematic
Q = 15.2200
b = 2.5100
C1 = 87468.3657
r1 = 2716.8016
C2 = 50462.7739
r2 = 368.7099
C3 = 2979.5208
r3 = 11.6512
C4 = 0.0000
r4 = 0.0000
Ignore bounds
100 %
l = 100
λ = 1×10⁻⁴
Idle
Visible
Run
Results
R²: 0.997312
RMSE: 19.709560
Export
Stress σ (MPa)
Strain ε (%)
31.00 32.00
0.00
755.50
Update
Show fit

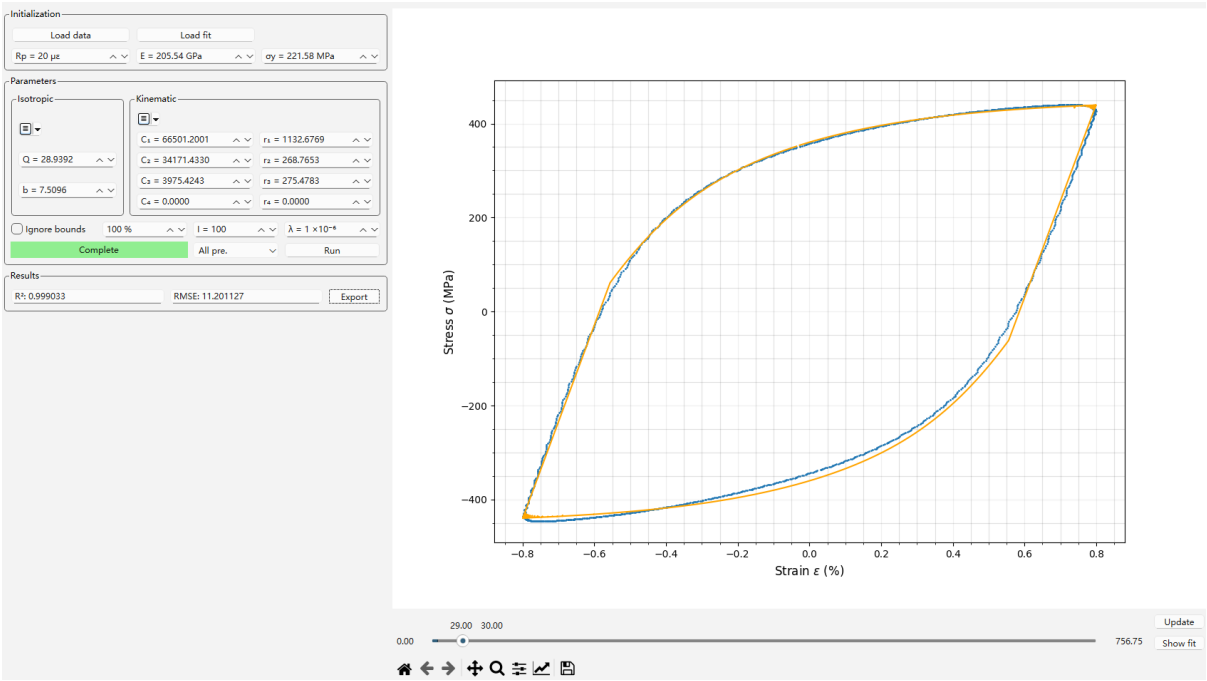
A_08am


Initialization
Load data
Load fit
Rp = 20 με
E = 181.08 GPa
σy = 199.04 MPa
Parameters
Isotropic
Kinematic
Q = 36.4155
b = 3.1000
C₁ = 122727.2677
C₂ = 80854.6818
C₃ = 16944.2364
C₄ = 0.0000
r₁ = 2984.1367
r₂ = 525.1518
r₃ = 18.0000
r₄ = 0.0000
Ignore bounds
100 %
l = 100
λ = 1 ×10⁻⁶
Complete
Visible
Run
Results
R²: 0.996384
RMSE: 22.381532
Export
Stress σ (MPa)
Strain ε (%)
32.00 33.00
0.00
779.00
Update
Show fit

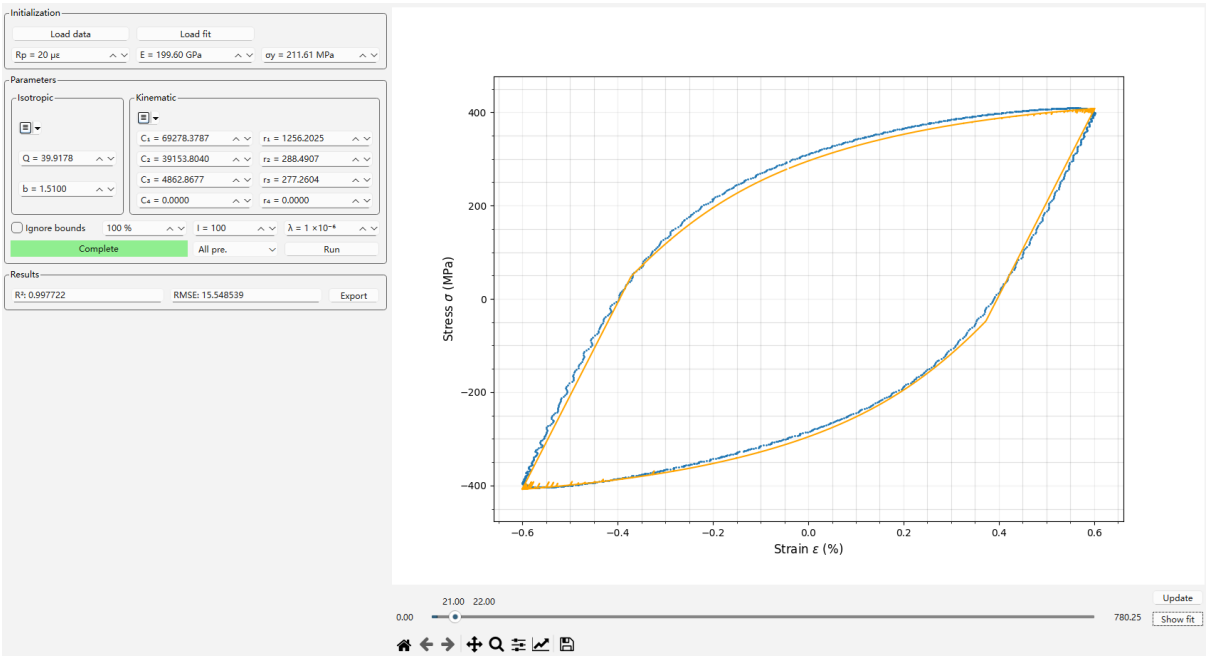
A_06am


Initialization
Load data
Load fit
Rp = 20 με
E = 207.20 GPa
σy = 229.96 MPa
Parameters
Isotropic
Kinematic
Q = 27.8665
b = 1.2965
C₁ = 55075.4901
C₂ = 33620.3431
C₃ = 3696.8970
C₄ = 0.0000
r₁ = 1089.3293
r₂ = 209.6962
r₃ = 194.5814
r₄ = 0.0000
Ignore bounds
100 %
l = 100
λ = 1 ×10⁻⁶
Complete
All pre.
Run
Results
R²: 0.998462
RMSE: 14.597248
Export
Stress σ (MPa)
Strain ε (%)
22.00 23.00
0.00
444.75
Update
Show fit

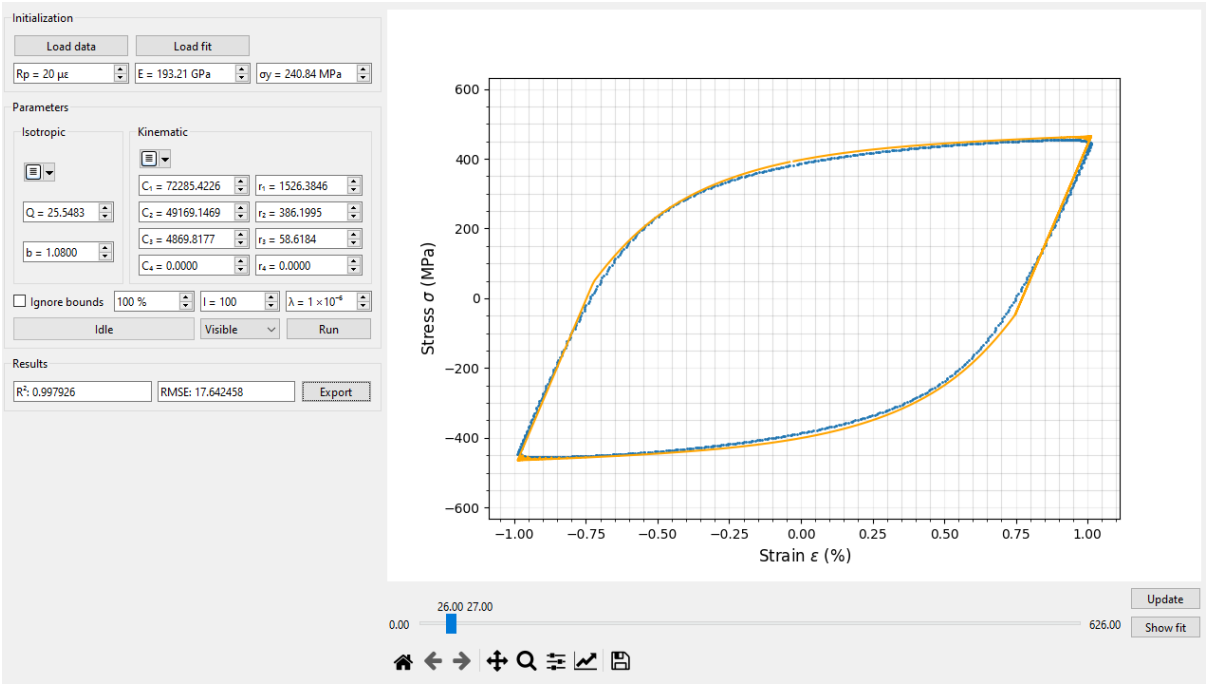
B_1am


Initialization
Load data
Load fit
Rp = 20 με
E = 205.54 GPa
σy = 221.58 MPa
Parameters
Isotropic
Q = 28.9392
b = 7.5096
Kinematic
C₁ = 66501.2001
C₂ = 34171.4330
C₃ = 3975.4243
C₄ = 0.0000
r₁ = 1132.6769
r₂ = 268.7653
r₃ = 275.4783
r₄ = 0.0000
Ignore bounds
100 %
I = 100
λ = 1 ×10⁻⁴
Complete
All pre.
Run
Results
R²: 0.999033
RMSE: 11.201127
Export
Stress σ (MPa)
Strain ε (%)
29.00
30.00
0.00
756.75
Update
Show fit

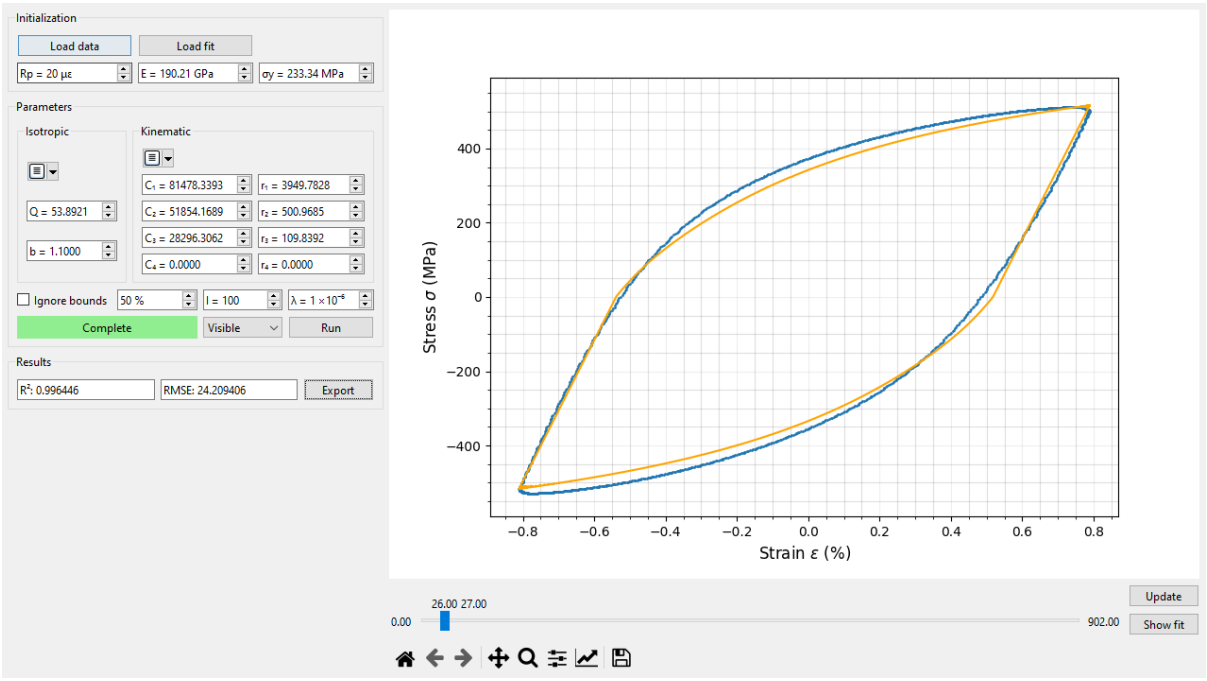
B_08am


Initialization
Load data
Load fit
Rp = 20 με
E = 199.60 GPa
σy = 211.61 MPa
Parameters
Isotropic
Q = 39.9178
b = 1.5100
Kinematic
C₁ = 69278.3787
C₂ = 39153.8040
C₃ = 4862.8677
C₄ = 0.0000
r₁ = 1256.2025
r₂ = 288.4907
r₃ = 277.2604
r₄ = 0.0000
Ignore bounds
100 %
I = 100
λ = 1 ×10⁻⁴
Complete
All pre.
Run
Results
R²: 0.997722
RMSE: 15.548539
Export
Stress σ (MPa)
Strain ε (%)
21.00
22.00
0.00
780.25
Update
Show fit

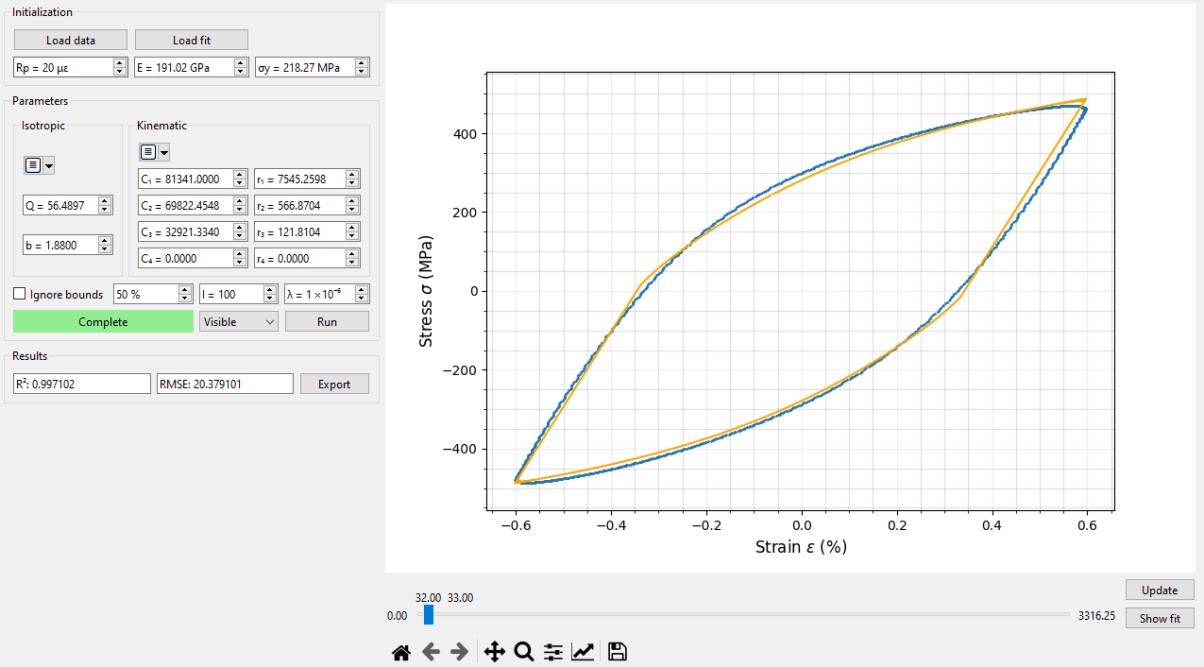
B_06am


Initialization
Load data
Load fit
Rp = 20 με
E = 193.21 GPa
σy = 240.84 MPa
Parameters
Isotropic
Kinematic
Q = 25.5483
b = 1.0800
C₁ = 72285.4226
r₁ = 1526.3846
C₂ = 49169.1469
r₂ = 386.1995
C₃ = 4869.8177
r₃ = 58.6184
C₄ = 0.0000
r₄ = 0.0000
Ignore bounds
100 %
l = 100
λ = 1 ×10⁻⁶
Idle
Visible
Run
Results
R²: 0.997926
RMSE: 17.642458
Export
Stress σ (MPa)
Strain ε (%)
26.00 27.00
0.00
626.00
Update
Show fit

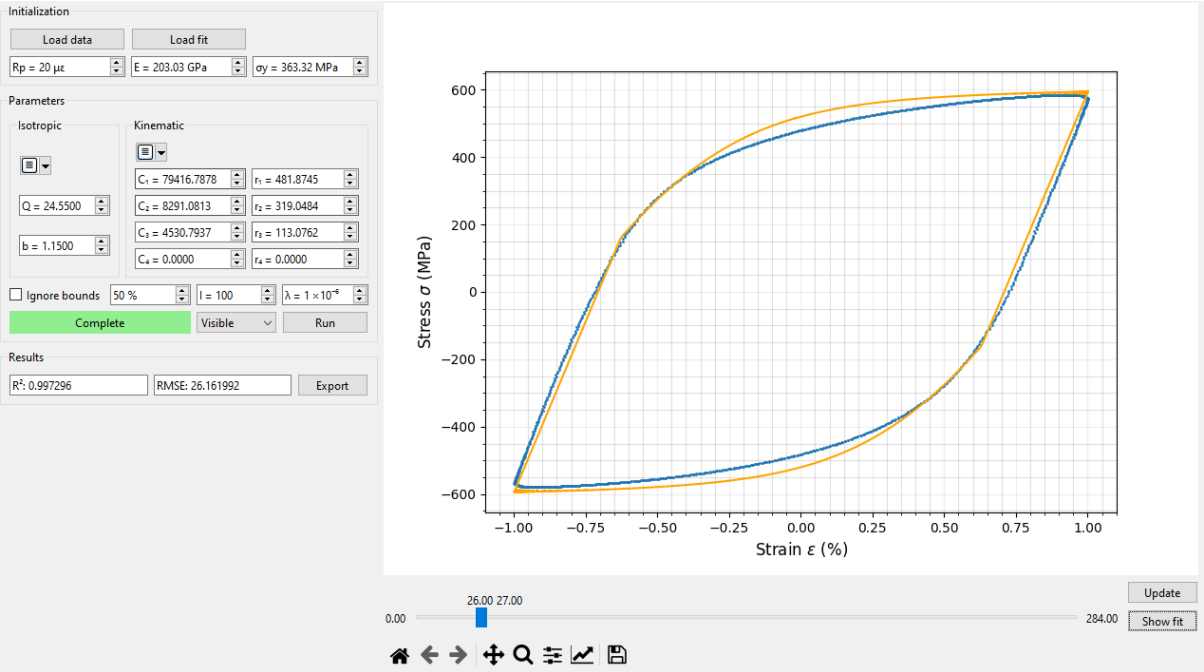
C_1am


Initialization
Load data
Load fit
Rp = 20 με
E = 190.21 GPa
σy = 233.34 MPa
Parameters
Isotropic
Kinematic
Q = 53.8921
b = 1.1000
C₁ = 81478.3393
r₁ = 3949.7828
C₂ = 51854.1689
r₂ = 500.9685
C₃ = 28296.3062
r₃ = 109.8392
C₄ = 0.0000
r₄ = 0.0000
Ignore bounds
50 %
l = 100
λ = 1 ×10⁻⁶
Complete
Visible
Run
Results
R²: 0.996446
RMSE: 24.209406
Export
Stress σ (MPa)
Strain ε (%)
26.00 27.00
0.00
902.00
Update
Show fit

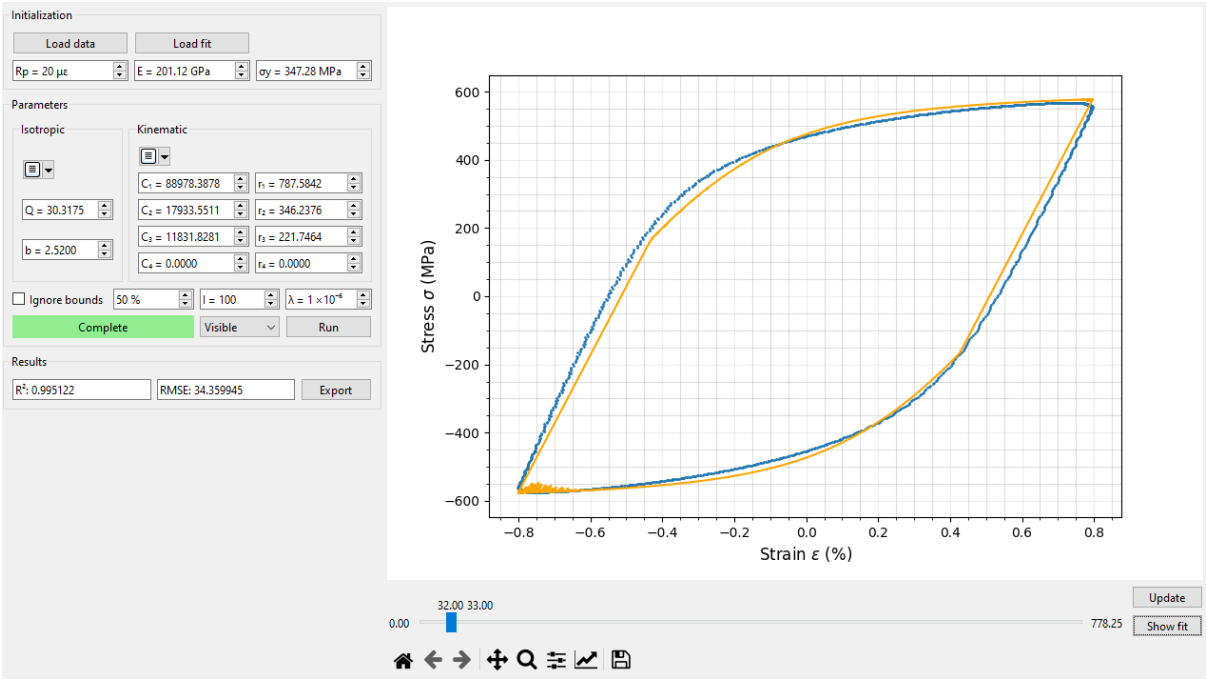
C_08am



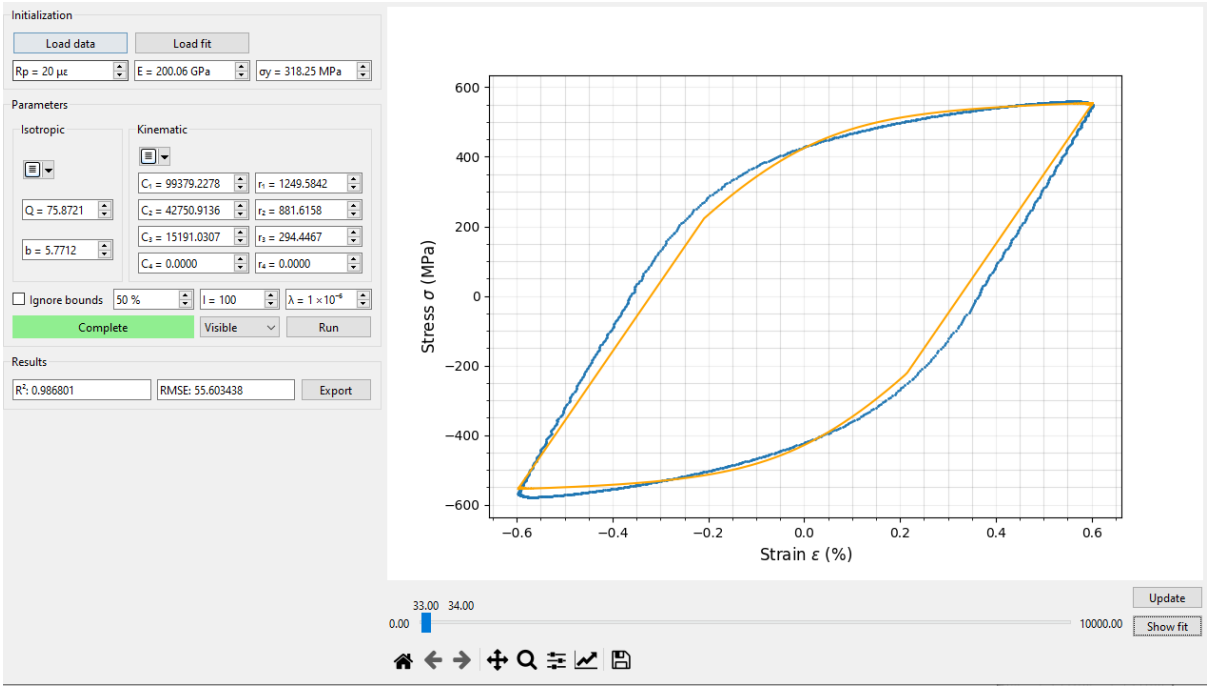
C_06am



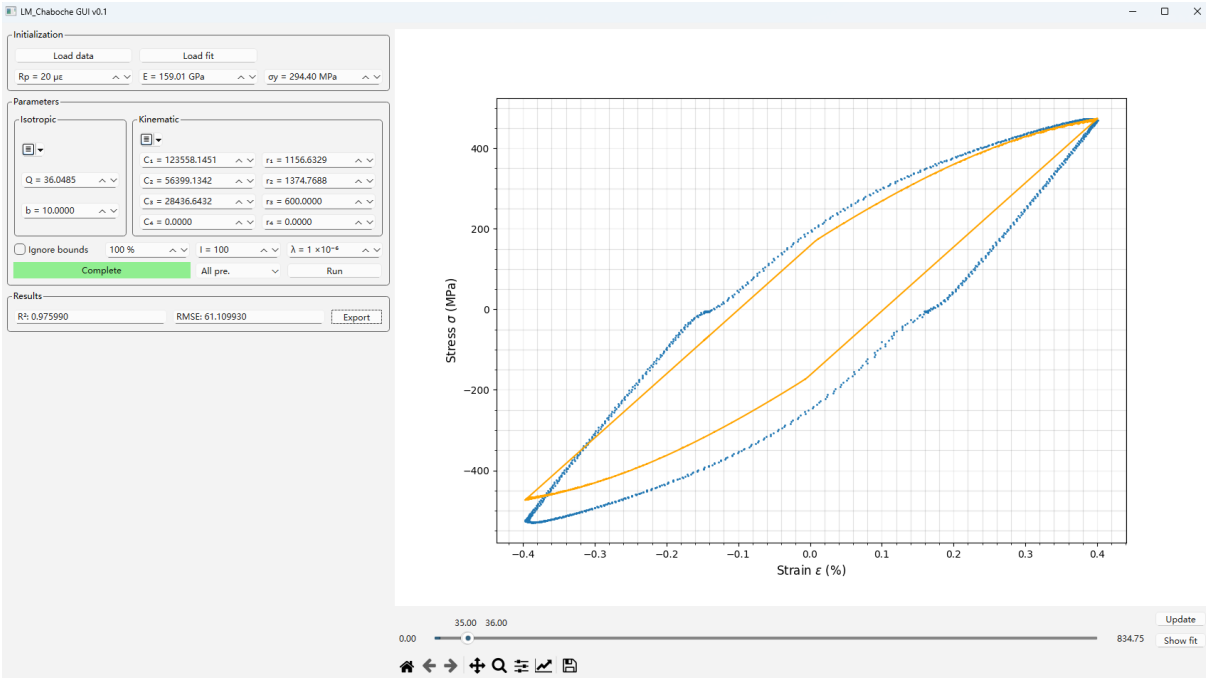
AM_H_1am



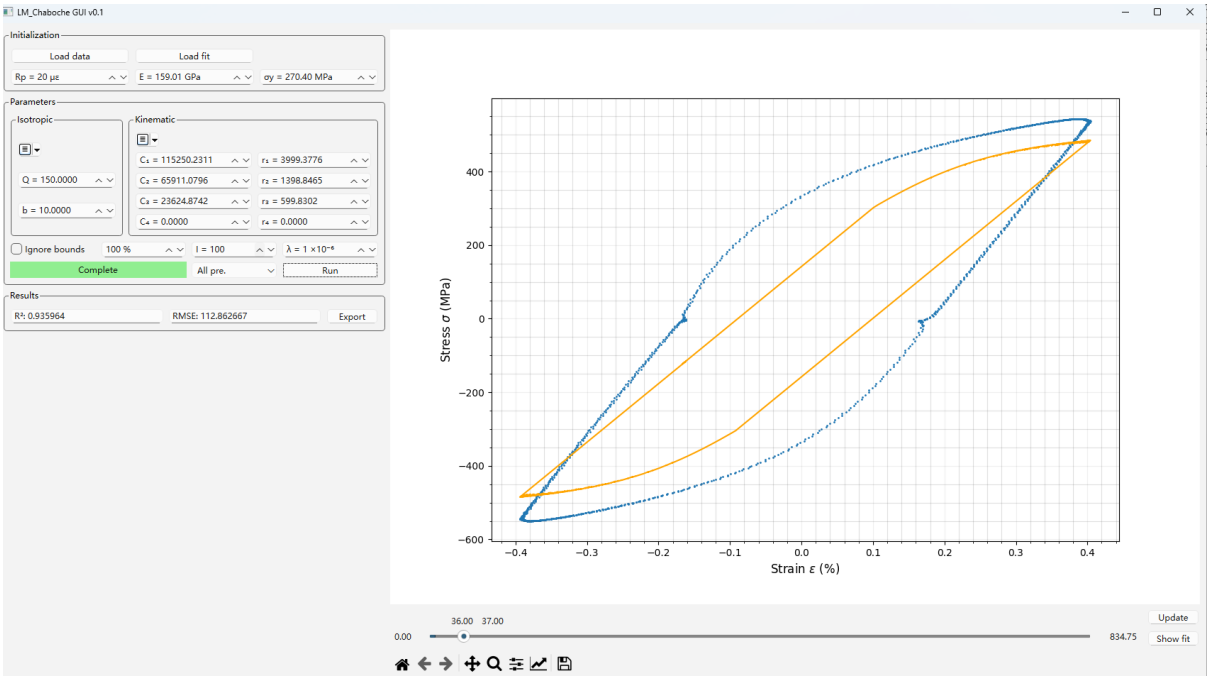
AM_H_08am



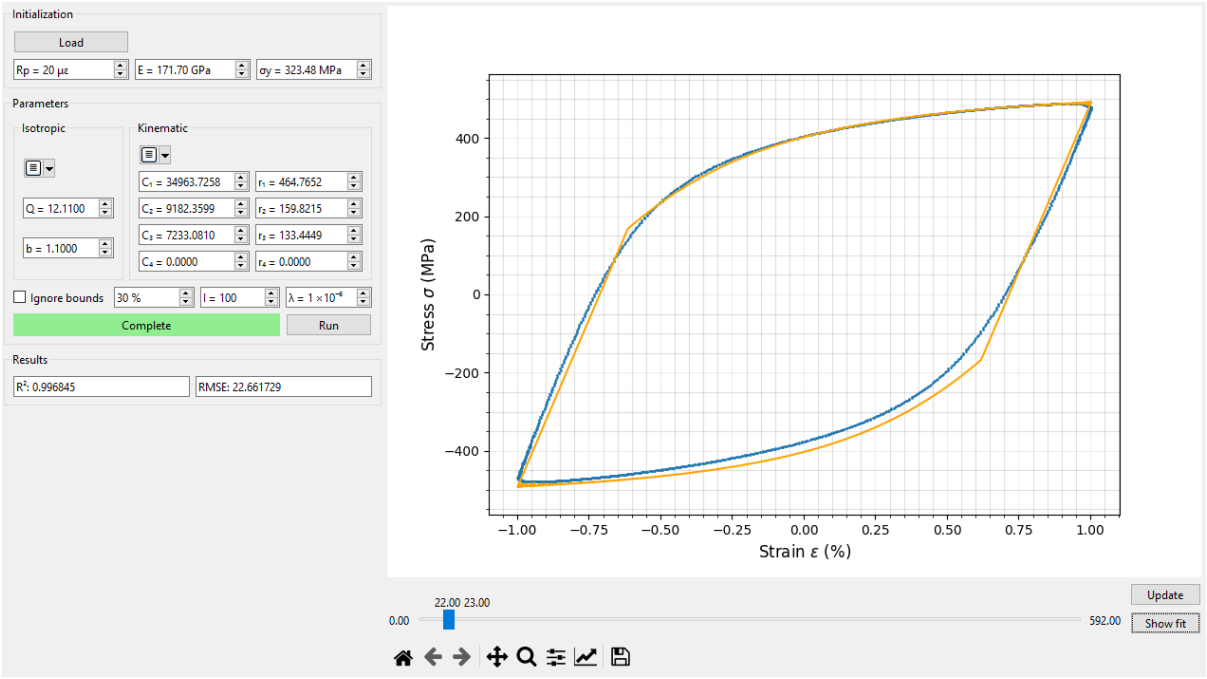
AM_H_06am



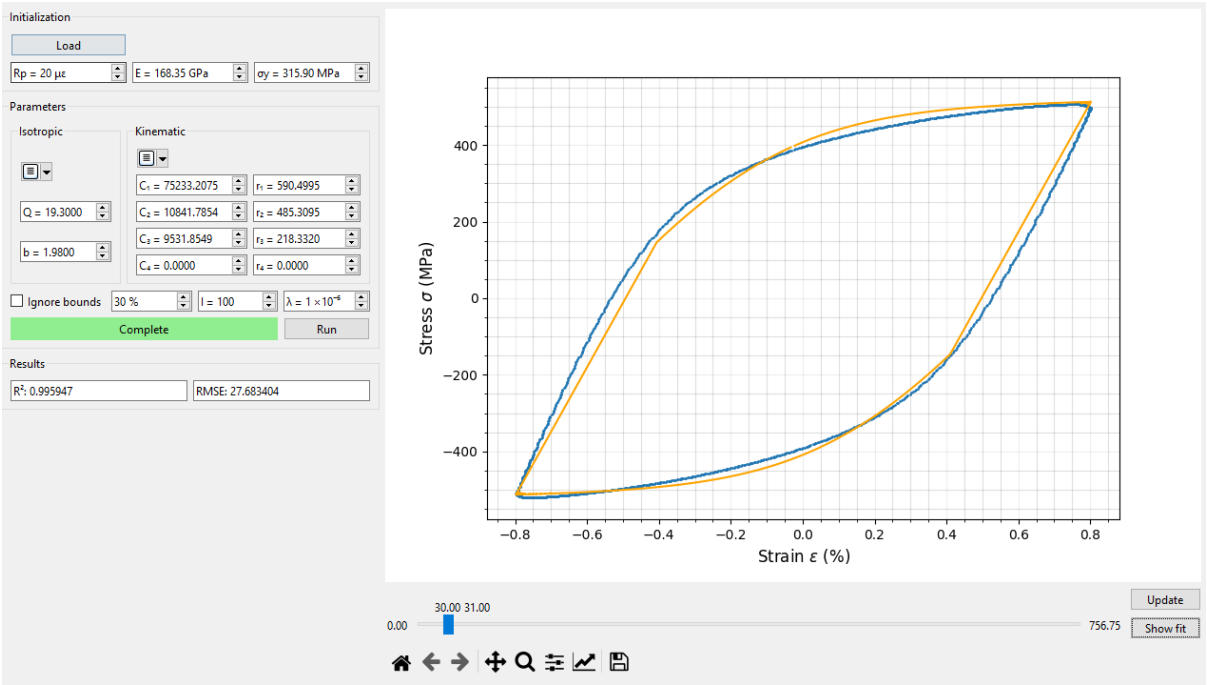
AM_V_04am



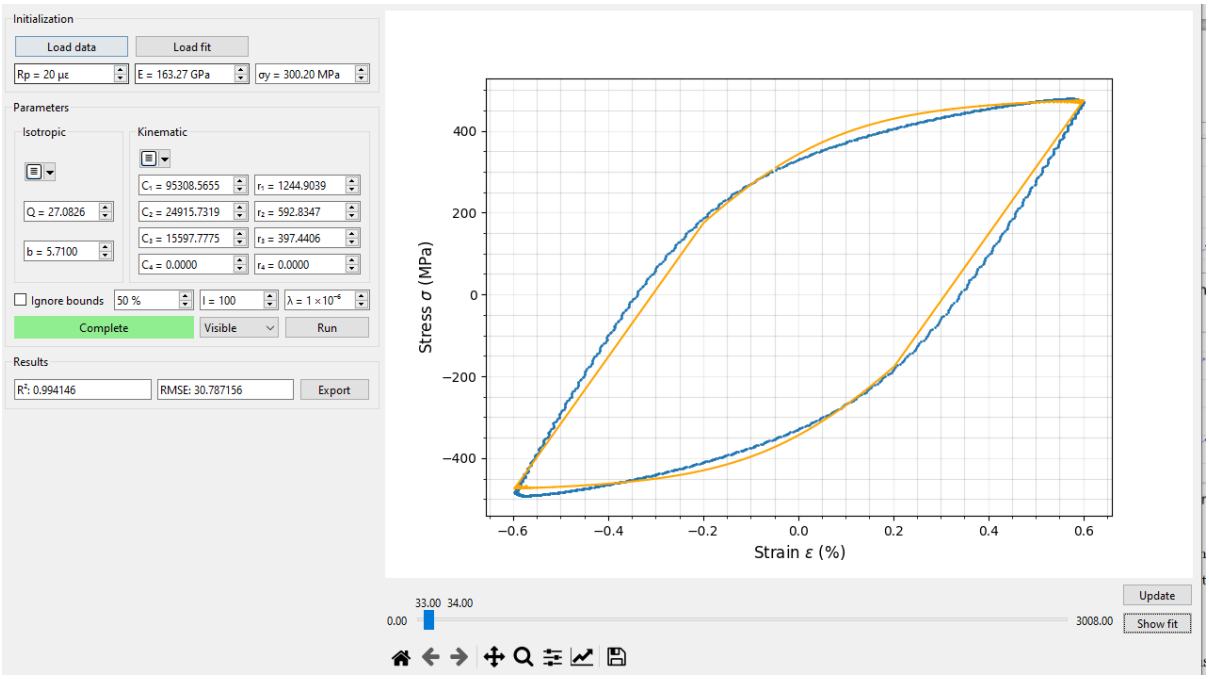
AM_H_04am



AM_V_1am



AM_V_08am


Initialization
Load data
Load fit
Rp = 20 με
E = 163.27 GPa
σy = 300.20 MPa
Parameters
Isotropic
Kinematic
Q = 27.0826
b = 5.7100
C1 = 95308.5655
C2 = 24915.7319
C3 = 15597.7775
C4 = 0.0000
r1 = 1244.9039
r2 = 592.8347
r3 = 397.4406
r4 = 0.0000
Ignore bounds
50 %
l = 100
λ = 1×10⁻⁶
Complete
Visible
Run
Results
R²: 0.994146
RMSE: 30.787156
Export
Stress σ (MPa)
Strain ε (%)
400
200
0
−200
−400
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
33.00
34.00
0.00
3008.00
Update
Show fit

AM_V_06am